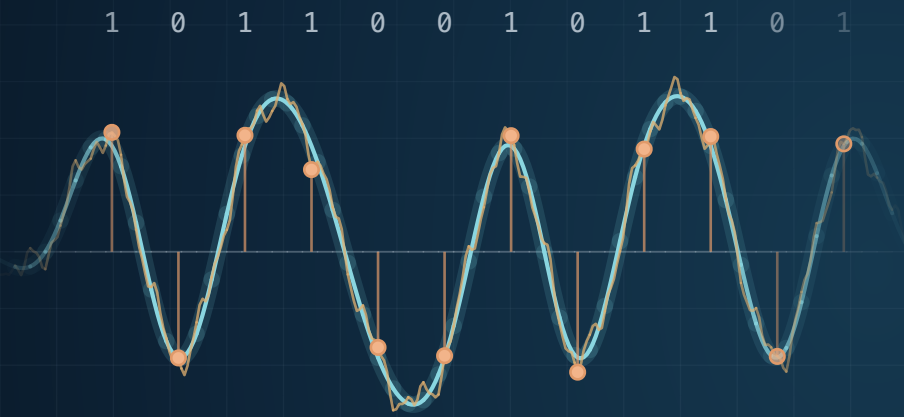




SAMPLING · DETECTION · MODULATION · CODING

Digital Communications

Formula and Notation Reference



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CONVENTIONS · FORMULAS · NOTATION

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Contents

The conventions used throughout the course, every formula it establishes, and every symbol it defines. Nothing here is derived; the derivations are in the lecture notes.

1	Conventions	3
	Energy and power, noise, the Q function, sinc, logarithms, and the two expressions every error probability comes from.	
2	Summary of formulas	5
	Everything the course establishes, in the order it establishes it.	
A.1	From an analog signal to bits — Chapter 1	5
A.2	Baseband transmission — Chapter 2	6
A.3	Signal space — Chapter 3	6
A.4	The optimal receiver — Chapter 4	7
A.5	Digital modulation — Chapter 5	7
A.6	Information theory — Chapter 6	8
3	Notation	9
	Every symbol the course defines.	

How to read it

Part 1 states the conventions, the two expressions every error probability comes from, and the transform pair Chapters 1 and 2 use. Part 2 is the summary of formulas, in the order the course establishes them. Part 3 defines every symbol. Nothing here is derived: where a result needs an argument, the argument is in the lecture notes chapter named beside it.

PART

1

Conventions

These hold everywhere in the course, without local variation. Half the factor-of-two errors in this subject come from mixing two of them in one line, so where a result depends on one it is restated at that point rather than assumed.

CONVENTION STATEMENT

Energy and power	Normalised: the resistance is taken as 1Ω , so instantaneous power is $ x(t) ^2$ and energy is its integral. Signal energy is E , energy per bit E_b , energy per symbol E_s , and $E_s = (\log_2 M) E_b$.
Noise	White and Gaussian with two-sided power spectral density $S_n(f) = N_0/2$. Projected onto an orthonormal basis it gives independent components, each $\mathcal{N}(0, N_0/2)$.
The Q function	$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$. It is the tail of a standard normal, so $Q(0) = \frac{1}{2}$ and $Q(-x) = 1 - Q(x)$.
sinc	Normalised: $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$, with zeros at every non-zero integer. Stated again wherever it is used.
Logarithms	log without a base means base two, and every entropy and every bit count is in bits. Decibels are $10 \log_{10}$ of a power or energy ratio.
Signals	$s_i(t)$ is transmitted, $r(t)$ is received, $\{\psi_k(t)\}$ is the orthonormal basis, and boldface $\mathbf{s}_i$ is the point in signal space.
Priors	Symbols are equally likely unless a problem says otherwise, in which case the MAP form is used and the prior term written out.

THE TWO EXPRESSIONS THE WHOLE COURSE RESTS ON

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

The first is exact for two signal points and is where every error probability in Chapters 2, 4 and 5 comes from. The second is its nearest-neighbour extension to M points, and the 2 in the denominator is the noise variance per dimension being $N_0/2$.

THE FOURIER TRANSFORM PAIR, AS CHAPTERS 1 AND 2 USE IT

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$

Written in hertz rather than in rad/s, so no factor of 2π appears in front of either integral. Sampling and the Nyquist criterion are both stated in this form.

PART

2

Summary of formulas

Everything the course establishes, in the order it establishes it. Nothing here is derived — where a result needs an argument, the argument is in the chapter named beside it.

A.1 From an analog signal to bits — Chapter 1

SAMPLING AND RECONSTRUCTION

$f_s > 2W$ (the sampling theorem, W the highest frequency present)

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Below the Nyquist rate the replicas of the spectrum overlap and the overlap cannot be undone. $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ throughout this course.

UNIFORM QUANTIZATION

$$\Delta = \frac{2V}{L} = \frac{2V}{2^n}, \quad |e| \leq \frac{\Delta}{2}$$

$$\sigma_q^2 = \frac{\Delta^2}{12}, \quad \operatorname{SQNR}|_{\text{dB}} = 6.02n + 1.76$$

The 1.76 assumes a full-scale sinusoid. The formula is an approximation at small n . At three bits it gives 19.82 dB where the measured value is 19.09 dB, and the gap halves with each extra bit.

Companding and the PCM bit rate.

RESULT	STATEMENT	CHAPTER
Companding	μ -law and A-law compress before quantizing so that the signal-to-noise ratio is flat across the range	1
PCM rate	$R = n f_s$ bits a second for n bits a sample	1

A.2 Baseband transmission — Chapter 2

THE MATCHED FILTER

$$h(t) = s(T - t), \quad \left(\frac{S}{N}\right)_{\max} = \frac{2E}{N_0}$$

The matched filter maximizes the signal-to-noise ratio at the sampling instant. The maximum depends on pulse energy, not pulse shape.

BINARY ERROR PROBABILITY AT BASEBAND

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \text{ (antipodal),} \quad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \text{ (on-off, orthogonal)}$$

NYQUIST CRITERION FOR ZERO INTERSYMBOL INTERFERENCE

$$\sum_k P\left(f + \frac{k}{T}\right) = T \quad \text{for all } f$$

$$B = \frac{1 + \alpha}{2T}, \quad 0 \leq \alpha \leq 1$$

The raised cosine meets the criterion for every roll-off α . At $\alpha = 0$ the bandwidth is the Nyquist minimum $1/2T$ and the pulse decays slowly. At $\alpha = 1$ it is twice that and the pulse decays quickly.

A.3 Signal space — Chapter 3

A SIGNAL AS A POINT

$$s_i(t) = \sum_{k=1}^N s_{ik} \psi_k(t), \quad s_{ik} = \int_0^T s_i(t) \psi_k(t) dt$$

$$E_i = \|\mathbf{s}_i\|^2 = \sum_{k=1}^N s_{ik}^2, \quad d_{ij} = \|\mathbf{s}_i - \mathbf{s}_j\|$$

The basis $\{\psi_k\}$ is orthonormal and comes from the Gram–Schmidt procedure applied to the signal set. Energy becomes a squared length and difference becomes a distance, which is the move the rest of the course rests on.

A.4 The optimal receiver — Chapter 4

THE DECISION RULE

$$\hat{s} = \arg \min_i \left\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right\} \quad (\text{MAP})$$

$$\hat{s} = \arg \min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad (\text{ML, equal priors})$$

With equal priors the rule is minimum distance. The equivalent correlation form is to maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$.

ERROR PROBABILITY

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q \left(\sqrt{\frac{d_{kj}^2}{2N_0}} \right)$$

$$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q \left(\sqrt{\frac{d_{kj}^2}{2N_0}} \right), \quad P_e \approx N_{\min} Q \left(\sqrt{\frac{d_{\min}^2}{2N_0}} \right)$$

The first result is exact for two points. The second is the union bound. The third is its nearest-neighbor form. Here, $N_{\min}$ is the average number of nearest neighbors and can be noninteger. The loose bound needs only M and $d_{\min}$.

A.5 Digital modulation — Chapter 5

Minimum squared distance and number of nearest neighbours for each scheme.

SCHEME	$d_{\min}^2$	$N_{\min}$
BPSK	$4E_b$	1
BFSK, BASK	$2E_b$	1
M -PSK	$4E_s \sin^2(\pi/M)$	2
M -PAM	$12E_s/(M^2 - 1)$	$2(M - 1)/M$
M -QAM (square)	$6E_s/(M - 1)$	3 at $M = 16$, 3.5 at $M = 64$
M -FSK	$2E_s$ for every pair	$M - 1$

READING THE TABLE

$$P_e \approx N_{\min} Q \left(\sqrt{\frac{d_{\min}^2}{2N_0}} \right), \quad E_s = (\log_2 M) E_b$$

Convert once between symbol and bit energy. BFSK and BASK need 3.01 dB more than BPSK for the same error probability. Doubling M costs approximately 6 dB in PAM. Moving from 4-PSK to 8-PSK costs 5.33 dB. At the same energy, 16-QAM gains 9.29 dB over 16-PAM.

A.6 Information theory — Chapter 6

INFORMATION AND ENTROPY

$$I(s_k) = -\log_2 p_k, \quad H(S) = -\sum_{k=1}^K p_k \log_2 p_k$$

$$0 \leq H(S) \leq \log_2 K, \quad H(S^n) = n H(S)$$

The upper bound is reached when all K symbols are equally likely. The extension result holds because the source is memoryless.

SOURCE CODING

$$\bar{L} = \sum_{k=1}^K p_k l_k, \quad \eta = \frac{H(S)}{\bar{L}} \leq 1$$

$$\sum_{k=1}^K 2^{-l_k} \leq 1, \quad H(S) \leq \bar{L} < H(S) + 1$$

The Kraft inequality is necessary but not sufficient for a prefix code. Over the n -th extension the upper bound becomes $H(S) + 1/n$, so $L_n/n \rightarrow H(S)$.

HUFFMAN CODES

$$\sigma^2 = \sum_{k=1}^K p_k (l_k - \bar{L})^2$$

Huffman is optimal in average length and is not unique. Placing a merged symbol as high as possible on a tie gives the minimum-variance code among the optimal ones.

PART

3

Notation

Every symbol the course defines, with the chapter that defines it. A symbol is never reused for a second meaning.

f_s, T_s

Sampling rate in hertz and the interval between samples, $T_s = 1/f_s$.

W

The highest frequency present in a message. The sampling theorem asks for $f_s > 2W$.

$\text{sinc}(x)$

Normalised: $\sin(\pi x)/(\pi x)$, with zeros at every non-zero integer.

Δ, L, n

Quantizer step, number of levels and bits a sample, with $L = 2^n$ and $\Delta = 2V/L$ over a range $\pm V$.

σ_q^2

Quantization noise power, $\Delta^2/12$ for a uniform quantizer.

T, R

Symbol interval and the rate that follows from it. A bit interval is written T_b and a bit rate R_b .

$p(t), h(t)$

The transmitted pulse and the impulse response of the receive filter. The matched filter is $h(t) = p(T - t)$.

α

Roll-off of a raised-cosine pulse, from 0 to 1. Bandwidth is $(1 + \alpha)/2T$.

$\psi_k(t)$

The k -th orthonormal basis function, from the Gram–Schmidt procedure applied to the signal set.

$s_i(t), \mathbf{s}_i$

The i -th transmitted waveform, and the point in signal space whose coordinates are its projections onto the basis.

E, E_i

Signal energy, normalised so that no resistance appears: $E_i = \|\mathbf{s}_i\|^2$.

$d_{ij}, d_{\min}$

Distance between two signal points, and the smallest such distance in a constellation.

$r(t), \mathbf{r}$

The received waveform and the observation vector it produces, $\mathbf{r} = \mathbf{s}_i + \mathbf{n}$.

N_0

Noise power spectral density parameter. The two-sided density is $N_0/2$ watts per hertz, and each noise component is $\mathcal{N}(0, N_0/2)$.

$Q(x)$

The Gaussian tail: the probability that a standard normal variable exceeds x . Equal to $\frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$.

 M, N

The number of signals in the set, and the number of basis functions they need. M signals carry $\log_2 M$ bits a symbol.

 $N_{\min}$

The average number of signal points at the minimum distance. An average, so not usually a whole number.

 P_b, P_e, P_s

Bit error probability, error probability in general, and symbol error probability.

 E_b, E_s

Energy per information bit and energy per transmitted symbol, with $E_s = (\log_2 M) E_b$.

 f_c

Carrier frequency. Every scheme in Chapter 5 changes the amplitude, the phase or the frequency of $\cos(2\pi f_c t)$.

 S, K, p_k

The source alphabet, how many symbols it has, and the probability of the k -th.

 $I(s_k)$

Self-information of one symbol, $-\log_2 p_k$ bits.

 $H(S)$

Entropy: the average information a symbol carries, in bits a symbol. Bounded by 0 and $\log_2 K$.

 $\bar{L}, l_k$

Average codeword length and the length of the codeword for symbol k .

 η

Coding efficiency, $H(S)/\bar{L}$. Never above one.

 σ^2

Variance of the codeword length about $\bar{L}$. Separates two Huffman codes that share an average.

 $p(y_k | x_j)$

Transition probability of a discrete memoryless channel. Collected into the channel matrix, whose every row sums to one.

 $H(X | Y)$

Conditional entropy: the uncertainty still remaining about the input after the output has been seen.

 $I(X; Y)$

Mutual information, $H(X) - H(X | Y)$: what the channel actually delivered, in bits per use. Symmetric and never negative.

 C

Channel capacity, the largest $I(X; Y)$ over all input distributions. For the binary symmetric channel, $1 - H(p)$.

 $H(p)$

The binary entropy function, $-p \log_2 p - (1 - p) \log_2 (1 - p)$. It measures a two-symbol source and also the noise a binary symmetric channel adds.

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VERSION	DATE	CHANGES
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